



# IndianOil

## JOURNEY RISK MANAGEMENT (JRM) STUDY

### Salem Terminal TO LINGAM ENTERPRISES

#### Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

#### Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

#### Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

## Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

## Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

## Environmental Considerations

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

## Journey Risk Mitigation

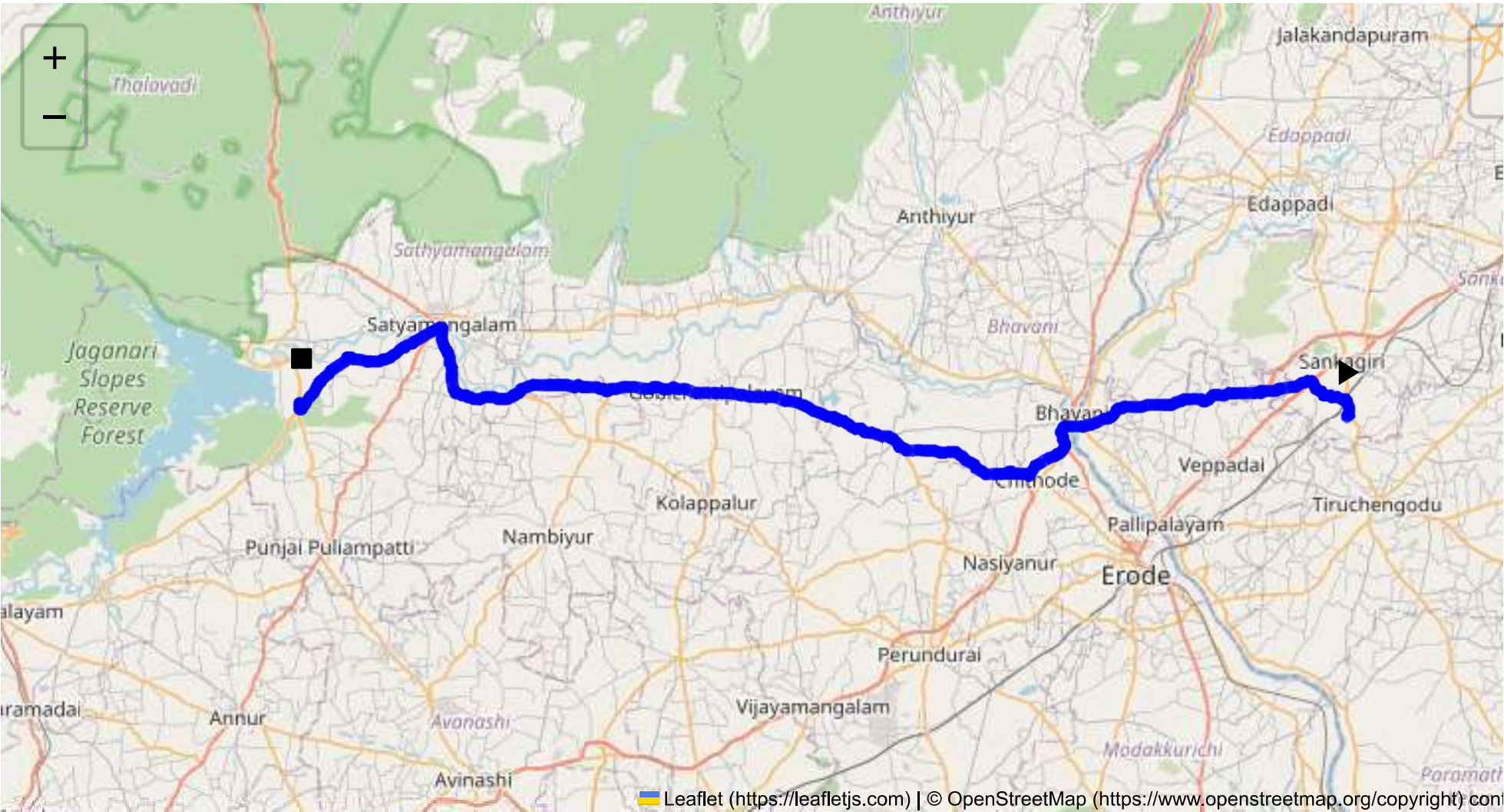
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

## Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



**Route Summary:**  
**Total Distance: 94.11 km**  
**Estimated Duration: 1.9 hours**  
**Adjusted Duration (Heavy Vehicle): 2.4 hours**  
**Start: (11.4381, 77.8734)**  
**End: (11.448338, 77.146328)**



Welcome to the Journey Risk Management Study

To conduct a comprehensive analysis of the route from CVQF+23W, Sangagiri, Tamil Nadu, India to C4XW+8G Nalroad, Tamil Nadu, for truck drivers transporting hazardous materials, here is a detailed report addressing the points requested:

1. Overview of the Route Map:

- The route spans approximately 94.11 kilometers and takes about 1.95 hours. The most direct path likely involves primary state highways and possibly sections of national highways. The journey will typically proceed southeast through rural and semi-urban areas.

## 2. Typical Weather Conditions and Potential Hazards:

- Tamil Nadu generally experiences a tropical climate. Weather conditions vary seasonally, with the northeast monsoon from October to December bringing heavy rains that can flood roads and reduce visibility. During the summer months (March to June), high temperatures can lead to road surface deterioration.

## 3. Traffic Patterns:

- Traffic is relatively light in rural areas but can be congested near towns and intersections, particularly during morning and evening rush hours (7-10 AM and 5-8 PM). Market days in towns increase congestion. Vehicle density typically heightens near educational institutions and industrial areas.

## 4. Assessment of Road Quality and Infrastructure:

- Roads are a mix of well-maintained highways and some secondary roads that may have potholes or narrow bridges. Attention should be paid to sections undergoing construction or repair, which are common after monsoon season.

## 5. Suggestions for Alternative Routes during Emergencies:

- National highways, where available, are preferable for detours due to better infrastructure. Ensure local alternate rural roads are pre-assessed for suitability, as they might be narrower or less maintained.

## 6. Local Regulations Affecting Hazardous Material Transport:

- Transporting hazardous materials requires adherence to specific permits in Tamil Nadu. Regulations dictate transit times and restrict routes through sensitive areas. Compliance with the Rules on the Transportation of Dangerous Goods in India is essential.

## 7. Historical Incidents Involving Heavy Vehicles or Hazardous Materials:

- While specific incidents on this exact route may not be well-documented, typical historical risks include vehicle overturns in sharp curves on rural roads and spillage due to road bumps or sudden braking.

## 8. Environmental Considerations and Sensitive Areas:

- Several agricultural zones and minor water bodies may be encountered along the route. Care should be taken to prevent contamination from leaks or spills, particularly near waterways and fertile lands.

## 9. Communication Coverage:

- Mobile network coverage is generally good along national highways but may become patchy on less-traveled rural roads. Dead zones are more likely near remote farmlands or in hilly terrain.

## 10. Estimated Emergency Response Times:

- In urban or semi-urban areas, emergency response times range from 15 to 30 minutes. Rural areas may experience delays of 45 minutes to an hour due to infrastructure and distance.

11. Overall Summary of Risk Assessment:

- The route is generally navigable with typical rural road challenges compounded by weather-related and infrastructure limitations. Key risk factors include road conditions in monsoon seasons, traffic congestion at peak times and in specific towns, and the need to follow stringent regulations for hazardous material transport. Constant communication with traffic control and emergency services is advised. Careful planning and an awareness of alternative routes and emergency contacts are critical for safe transit.

Risk Assessment - Turns

|    | Risk Type  | Risk Level | Coordinates        | Speed Limit |
|----|------------|------------|--------------------|-------------|
| 0  | Roundabout | High       | 11.49613, 77.23785 | 15 KM/Hr    |
| 1  | Turn       | High       | 11.43968, 77.87345 | 15 KM/Hr    |
| 2  | Turn       | High       | 11.44024, 77.87528 | 15 KM/Hr    |
| 3  | Turn       | High       | 11.44898, 77.87410 | 15 KM/Hr    |
| 4  | Turn       | Medium     | 11.45357, 77.85789 | 30 KM/Hr    |
| 5  | Turn       | High       | 11.45352, 77.85698 | 15 KM/Hr    |
| 6  | Turn       | Medium     | 11.45847, 77.85140 | 30 KM/Hr    |
| 7  | Turn       | Medium     | 11.45898, 77.85135 | 30 KM/Hr    |
| 8  | Turn       | High       | 11.46303, 77.84945 | 15 KM/Hr    |
| 9  | Turn       | Medium     | 11.45509, 77.81383 | 30 KM/Hr    |
| 10 | Turn       | Medium     | 11.45085, 77.77020 | 30 KM/Hr    |
| 11 | Turn       | Medium     | 11.43165, 77.67734 | 30 KM/Hr    |
| 12 | Turn       | High       | 11.39829, 77.65233 | 15 KM/Hr    |
| 13 | Turn       | Medium     | 11.45763, 77.36309 | 30 KM/Hr    |
| 14 | Turn       | Medium     | 11.45750, 77.35617 | 30 KM/Hr    |
| 15 | Turn       | Medium     | 11.45345, 77.29440 | 30 KM/Hr    |
| 16 | Turn       | Medium     | 11.45536, 77.25423 | 30 KM/Hr    |
| 17 | Turn       | Medium     | 11.49069, 77.24436 | 30 KM/Hr    |
| 18 | Blind Spot | Blind Spot | 11.49929, 77.24500 | 10 KM/Hr    |
| 19 | Turn       | High       | 11.49933, 77.24485 | 15 KM/Hr    |
| 20 | Blind Spot | Blind Spot | 11.44360, 77.14632 | 10 KM/Hr    |

## Emergency Locations

|    | type     | name                                           | coordinates            | speed_limit | risk_level |
|----|----------|------------------------------------------------|------------------------|-------------|------------|
| 3  | hospital | Dhanvantri Multi Speciality Hospital           | 11.451362, 77.766602   | 30 km/h     | Medium     |
| 4  | hospital | Dhanvanthri Hospital                           | 11.4496712, 77.7593772 | 30 km/h     | Medium     |
| 5  | hospital | J.K.K. Trust Hospital                          | 11.4445841, 77.7307962 | 30 km/h     | Medium     |
| 7  | hospital | Shri Sathyanarayana Hospital                   | 11.4291297, 77.6913408 | 30 km/h     | Medium     |
| 8  | hospital | Thanish Siddha Hospital                        | 11.430003, 77.674964   | 30 km/h     | Medium     |
| 9  | clinic   | Harshitha Clinic                               | 11.4313207, 77.674718  | 30 km/h     | Medium     |
| 10 | hospital | Sri Kaalangi Siddhar Mooligai Vaithiya Nilayam | 11.432369, 77.674894   | 30 km/h     | Medium     |
| 11 | clinic   | G.K Clinic                                     | 11.4297244, 77.6749715 | 30 km/h     | Medium     |
| 12 | hospital | Naarayani Hospital                             | 11.4157401, 77.5773087 | 30 km/h     | Medium     |
| 13 | hospital | P K Hospital                                   | 11.4175923, 77.5640962 | 30 km/h     | Medium     |
| 14 | hospital | Dharun Hospital                                | 11.4223638, 77.5597735 | 30 km/h     | Medium     |
| 15 | hospital | Dr. Ramaswamy Hospital                         | 11.4249065, 77.5535658 | 30 km/h     | Medium     |
| 16 | hospital | Dakshin Trauma Centre                          | 11.454593, 77.4461819  | 30 km/h     | Medium     |
| 17 | hospital | Sri Varahi Herbal Research Centre and Hospital | 11.4539333, 77.4427438 | 30 km/h     | Medium     |
| 18 | hospital | Dakshin Hospital                               | 11.4546301, 77.445905  | 30 km/h     | Medium     |
| 19 | clinic   | Germen Homeopathy Clinic                       | 11.4536305, 77.4394692 | 30 km/h     | Medium     |
| 20 | hospital | JP Hospital                                    | 11.454765, 77.438257   | 30 km/h     | Medium     |
| 21 | hospital | KKS Hospital                                   | 11.4551792, 77.4392927 | 30 km/h     | Medium     |
| 23 | hospital | Siranjeeve Hospitals                           | 11.453132, 77.4344382  | 30 km/h     | Medium     |

|    | type     | name                           | coordinates               | speed_limit | risk_level |
|----|----------|--------------------------------|---------------------------|-------------|------------|
| 24 | clinic   | Anandhan Clinic                | 11.4548037,<br>77.4349074 | 30 km/h     | Medium     |
| 25 | hospital | Ramachandran Hospital          | 11.452242, 77.434641      | 30 km/h     | Medium     |
| 26 | hospital | Kumaran Hospital               | 11.4533971,<br>77.4349621 | 30 km/h     | Medium     |
| 27 | hospital | Dr. Rathinasamy Hospital       | 11.457562,<br>77.4327752  | 30 km/h     | Medium     |
| 28 | hospital | Gobi Neuro Mind Care           | 11.4548344,<br>77.434649  | 30 km/h     | Medium     |
| 29 | hospital | Hariharan Hospital             | 11.4523578,<br>77.4299287 | 30 km/h     | Medium     |
| 30 | hospital | Vasanth Hospital               | 11.4526864,<br>77.4292173 | 30 km/h     | Medium     |
| 31 | clinic   | Jeeva Nursing Home             | 11.4533031,<br>77.4294145 | 30 km/h     | Medium     |
| 32 | hospital | Sathya Hospital                | 11.4561856,<br>77.4298694 | 30 km/h     | Medium     |
| 33 | hospital | S K Hospital                   | 11.4540106,<br>77.430108  | 30 km/h     | Medium     |
| 34 | hospital | Sivakami Hospital              | 11.4531846,<br>77.4299288 | 30 km/h     | Medium     |
| 35 | clinic   | Sri Sudha Physiotherapy Clinic | 11.4559915,<br>77.4300217 | 30 km/h     | Medium     |
| 36 | hospital | Best Hospital                  | 11.4543558,<br>77.4263617 | 30 km/h     | Medium     |
| 37 | clinic   | Anand Nursing Home             | 11.4553169,<br>77.4296248 | 30 km/h     | Medium     |
| 38 | clinic   | Kanmani Child Care Clinic      | 11.4545273,<br>77.428668  | 30 km/h     | Medium     |
| 39 | hospital | Rajasekaran Hospital           | 11.454463, 77.4262        | 30 km/h     | Medium     |
| 40 | hospital | Samuel Hospital                | 11.4542555,<br>77.4271194 | 30 km/h     | Medium     |
| 41 | hospital | Sivasakthi Hospital            | 11.4544711,<br>77.4297229 | 30 km/h     | Medium     |
| 42 | hospital | SS Children's Hospital         | 11.4555386,<br>77.4296357 | 30 km/h     | Medium     |
| 43 | clinic   | Sri Dhanvantari Siddha Clinic  | 11.4561216,<br>77.4295797 | 30 km/h     | Medium     |

|    | type     | name                                       | coordinates               | speed_limit | risk_level |
|----|----------|--------------------------------------------|---------------------------|-------------|------------|
| 44 | hospital | Neha Hospital                              | 11.4560138,<br>77.4270219 | 30 km/h     | Medium     |
| 45 | hospital | M.S Children's Hospital                    | 11.4556012,<br>77.4301006 | 30 km/h     | Medium     |
| 46 | hospital | Gobi Diabetic Hospital                     | 11.455394,<br>77.4301181  | 30 km/h     | Medium     |
| 47 | clinic   | Dr. Ramesh Babu Clinic                     | 11.4560871,<br>77.4296353 | 30 km/h     | Medium     |
| 48 | hospital | Sankareswari Eye Hospital                  | 11.4572997,<br>77.428765  | 30 km/h     | Medium     |
| 49 | hospital | Arun Hospital                              | 11.4548852,<br>77.4250314 | 30 km/h     | Medium     |
| 52 | hospital | Government Hospital                        | 11.4565126,<br>77.4288768 | 30 km/h     | Medium     |
| 53 | hospital | Sri Chakra Hospital                        | 11.4544464,<br>77.4228363 | 30 km/h     | Medium     |
| 54 | hospital | U M Hospital                               | 11.4544151,<br>77.4240386 | 30 km/h     | Medium     |
| 55 | hospital | Dr. Gopal Hospital                         | 11.455258, 77.422901      | 30 km/h     | Medium     |
| 56 | clinic   | VG Ortho Centre                            | 11.4550504,<br>77.4225102 | 30 km/h     | Medium     |
| 57 | clinic   | Palaniappa Clinic                          | 11.4555925,<br>77.4202828 | 30 km/h     | Medium     |
| 58 | hospital | Abi SK Hospital                            | 11.457163, 77.412487      | 30 km/h     | Medium     |
| 59 | clinic   | Nalam Vazh S.K Siddha and Varma Clinic     | 11.457576, 77.412164      | 30 km/h     | Medium     |
| 61 | clinic   | Singiripalayam Natarajan Siddha Ortho Care | 11.4603252,<br>77.3183801 | 30 km/h     | Medium     |
| 63 | hospital | Pee Vee Hospital                           | 11.4960866,<br>77.2437224 | 30 km/h     | Medium     |
| 64 | hospital | Sakthi Hospital                            | 11.498927,<br>77.2433702  | 30 km/h     | Medium     |
| 65 | hospital | Anush Hospital                             | 11.500453,<br>77.2449535  | 30 km/h     | Medium     |

## Crowded Spots

|    | type        | name                                           | coordinates               | speed_limit | risk_level |
|----|-------------|------------------------------------------------|---------------------------|-------------|------------|
| 0  | school      | KRP Matric. Hr. Sec School                     | 11.4546193,<br>77.8142445 | 30 km/h     | Medium     |
| 1  | college     | Vivekanandha Engineering College               | 11.4589312,<br>77.7899284 | 30 km/h     | Medium     |
| 2  | marketplace | Monday market                                  | 11.452863,<br>77.775989   | 30 km/h     | Medium     |
| 6  | school      | SSM Matriculation Higher Secondary School      | 11.4321653,<br>77.6880046 | 30 km/h     | Medium     |
| 22 | school      | Shree Vidyalaya Matric Higher Secondary School | 11.4536114,<br>77.4383163 | 30 km/h     | Medium     |
| 50 | school      | Diamond Jubilee Higher Secondary School        | 11.4529541,<br>77.4280306 | 30 km/h     | Medium     |
| 51 | school      | CKK Palaniammal School                         | 11.4569211,<br>77.4276828 | 30 km/h     | Medium     |
| 60 | school      | Saratha Techno School                          | 11.4544819,<br>77.379986  | 30 km/h     | Medium     |
| 62 | school      | Saratha International School                   | 11.4541471,<br>77.3032962 | 30 km/h     | Medium     |

Route Photos of Risky Spots



Risk Type: Roundabout  
Risk Level: High  
Speed Limit: 15 KM/Hr  
Coordinates: 11.49613, 77.23785



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.44024, 77.87528



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.44898, 77.87410



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.45357, 77.85789



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.45352, 77.85698



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.45847, 77.85140



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.45898, 77.85135



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.46303, 77.84945



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.45509, 77.81383



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.45085, 77.77020



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.43165, 77.67734



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.39829, 77.65233



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.45763, 77.36309



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.45750, 77.35617



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.45345, 77.29440



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.45536, 77.25423



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.49069, 77.24436



**Risk Type:** Blind Spot  
**Risk Level:** Blind Spot  
**Speed Limit:** 10 KM/Hr  
**Coordinates:** 11.49929, 77.24500



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.49933, 77.24485



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.44360, 77.14632